A.5.1 Setting an exploration schedule for Upper Confidence Bound

This exercise, implemented in `CH05/02 - Exercise 1.ipynb`, discusses a strategy of adaptively setting the value of the tradeoff parameter β of the Upper Confidence Bound (UCB) policy. There are multiple steps:

1. Recreate the Bayesian optimization (BayesOpt) loop in `CH04/02 - Exercise 1.ipynb`, which uses the one-dimensional Forrester function as the optimization objective.
2. What is the value the multiplier for needs to have?

   Since there are ten iterations in the BayesOpt loop, is multiplied by the multiplier m ten times to go from 1 to 10. That is, $1\ m^{10} = 10$. Solving this equation gives the code for the multiplier:

   ```
   num_queries = 10

   start_beta = 1
   end_beta = 10

   multiplier = (end_beta / start_beta) ** (1 / num_queries)
   ```
3. Implement this scheduling logic and observe the resulting optimization performance.

   We modify the BayesOpt loop as follows:

   ```
   num_queries = 10

   start_beta = 1
   end_beta = 10

   multiplier = (end_beta / start_beta) ** (1 / num_queries)

   beta = start_beta

   for i in range(num_queries):
       ... ❶

       policy = botorch.acquisition.analytic.UpperConfidenceBound(
           model, beta=beta
       )

       ... ❷

       beta *= multiplier
   ```

   ❶ Obtain the trained Gaussian process.

   ❷ Find the point that maximizes the acquisition score, query the objective function, and update the training data.

   This code produces Figure A.10.

©Manning Publications Co. To comment go to liveBook
https://livebook.manning.com/#!/book/bayesian-optimization-in-action/discussion

Licensed to ashwin sampath <ashwin.manning@gmail.com>